NAGASAKI UNIVERSITY · TPO EXCHANGE PROGRAMME 2026

長崎大学 TPOインターンシップ 2026

Missing Data Theory & Ad-Hoc & Weighted Inference

欠測データ理論とアドホック・重み付け推論手法

A four-week research summary, presented as an exchange student

交換留学生として発表する4週間の研究概要

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AGENDA 本日の構成

What This Talk Covers

01

Academic Work

学術研究

Three connected strands of statistical research on missing data and valid inference, completed during the internship.

02

Cultural Experience

文化体験

Reflections from three cities explored during my time in Japan — Tokyo, Osaka, and Kyoto.

PART 1

Academic Work

学術研究

Missing-data theory, ad-hoc deletion, and weighted inference methods - Book Source (Missing Data Analysis and Design)

RESEARCH CONTEXT　研究の背景

Three Strands, One Question

「不完全なデータからでも、どうすれば正しい推論ができるか」

How do we draw valid conclusions from data that is incomplete, without silently distorting the variances and relationships hidden inside it?

Missing Data Theory

欠測データ理論

MCAR / MAR / NMAR mechanisms, single imputation, and the case for multiple imputation.

Ad-Hoc & Imputation

アドホック手法と重み付け・代入法

Complete-case, available-case, and weighted analysis, plus the full single-imputation family (mean through PMM) - what each method assumes, and what it costs in bias or variance.

Simulation Verification

シミュレーション検証

Simulated income/spending data test whether textbook bias and variance formulas actually hold.

FOUNDATIONS 基礎

Why Missing Data Matters

欠測データがなぜ重要なのか

A dataset is rarely perfect. Discarding incomplete rows wastes information; filling them in naively can quietly distort every downstream conclusion.

Means	Variances	Covariances	Regression Slopes	Standard Errors	p-values
平均	*分散*	*共分散*	*回帰係数*	*標準誤差*	*p値*

Key Takeaway

Single imputation can distort every one of these quantities at once. The chapter-long lesson: treat imputed values as predictions, never as real observations.

MCAR, MAR, and NMAR

3つの欠測メカニズム — なぜ重要なのか

MCAR

Missing Completely at Random

完全にランダムな欠測

Missingness is unrelated to any value — observed or missing. Complete-case analysis stays unbiased.

MAR

Missing at Random

観測情報で説明できる欠測

Missingness depends only on observed variables. Conditioning on them restores validity.

NMAR

Not Missing at Random

欠測値自体に依存する欠測

Missingness depends on the unobserved value itself, even after conditioning. The hardest case.

Explicit, Implicit & Hybrid Models

統計モデルを選ぶ3つのアプローチ

MCAR

Explicit Modeling

明示的モデリング

States the statistical model openly: means, regression, Buck's method. Assumptions are visible — but a wrong model gives systematically wrong values.

Implicit

Implicit Modeling

暗黙的モデリング

Uses an algorithm that implies a model: hot deck, cold deck, substitution. Every algorithm still carries hidden assumptions.

Hybrid

Predictive Mean Matching

予測平均マッチング（ハイブリッド）

Fits a model to define similarity, then imputes an actual donor's value — regression structure with hot-deck realism.

The Full Method Landscape

削除から多重代入まで— 学んだ手法のすべて

Four stages, one throughline: every method either discards information, reweights it, or predicts it. Each stage relaxes an assumption the previous stage needed.

01

Complete- & Available-Case

完全・利用可能ケース

Discard incomplete rows, or use every observed value pairwise.

02

Weighting Methods

重み付け手法

Reweight survivors by inverse response probability.

03

Single Imputation

単一代入法

Fill gaps: 10 methods, from a single mean to a matched donor.

04

Multiple Imputation

多重代入法

Impute M times, pool with Rubin's Rules - honest uncertainty.

Complete-Case Analysis

完全ケース分析— 単純さと引き換えの代償

Restrict the entire analysis to rows with no missing values on any variable used. The default in most software - and the baseline every other method is judged against.

ADVANTAGE

Simplicity & Comparability

Any standard complete-data method (regression, ANOVA) applies unmodified. Every statistic is computed on the same sample base, so cross-tabs stay mutually consistent.

DISADVANTAGE

Loss of Precision

Discarding rows reduces the effective sample size, increasing variance - even when missingness is completely random (MCAR).

DISADVANTAGE

Bias

If complete cases are not a random subsample (mechanism is not MCAR), CC estimates are biased for the population quantity of interest.

Quantifying the Loss of Precision

バイアスがなくても分散が増える理由— Δ_CC の定義

Bivariate normal (Y1,Y2), monotone missingness: Y1 observed for all n; Y2 observed for r of them. Data are MCAR, so this isolates precision loss from bias.

DEFINITION · Δ_CC = 1 means CC needs double the variance of the efficient estimator

$$\mathrm{Var}(\hat{\theta}_{\mathrm{CC}}) = \mathrm{Var}(\hat{\theta}_{\mathrm{EFF}})\,(1+\Delta_{\mathrm{CC}})$$

Mean of Y1

$$\boxed{\Delta_{\mathrm{CC}} = \frac{n-r}{r}}$$

CC discards the n-r rows with Y2 missing, even though their Y1 values are perfectly good. Wasteful - if r=n/2, variance doubles.

Mean of Y2

$$\boxed{\Delta_{\mathrm{CC}} \approx \frac{(n-r)\rho^2}{n(1-\rho^2)+r\rho^2}}$$

CC loses the n-r rows outright, and fails to use the correlated Y1 information for them. As $\rho^2 \to 1$, this converges to (n-r)/r.

Slope of Y2 on Y1

$$\boxed{\Delta_{\mathrm{CC}} = 0}$$

Fully efficient. Cases with Y1 observed but Y2 missing carry no information about Y2|Y1 - that relationship is only learned where both are observed.

Two Proofs Worth Remembering

完全ケース分析における2つの理論的結果

BIAS OF THE CC MEAN

Bias = (missing fraction) × (respondent/non-respondent gap)

$$\boxed{\mu_{CC} - \mu = (1 - p_{CC})(\mu_{CC} - \mu_{IC})}$$

Numerical example: 30% refuse to disclose income, and refusers earn $20k more on average. Bias = 0.3×(−20) = −6: CC understates population income by $6,000.

CC regression coefficients stay unbiased provided Pr(M=1|X,Y) = Pr(M=1|X) — missingness may depend on covariates any way, but not further on Y once X is conditioned upon.

ODDS-RATIO INVARIANCE

$$\mathrm{OR}_{\mathrm{CC}} = \mathrm{OR}$$

$$\boxed{\log \Pr(M=1 \mid Y_1, Y_2) = \alpha + g_1(Y_1) + g_2(Y_2)}$$

If response probability factors additively (row effect × column effect, no interaction), the complete-case odds ratio equals the population odds ratio exactly — proof by cancelling the nuisance row/column factors in numerator and denominator.

This is the formal justification for case-control sampling in epidemiology: sampling diseased and healthy people at very different rates is exactly such a row effect, so the exposure odds ratio remains valid.

Fixing Bias With Weights

重み付けクラスと傾向スコア

WEIGHTING CLASS ESTIMATOR

$$w_i = \frac{n_j}{r_j}$$

$$\boxed{\bar{y}_{\mathrm{wc}} = n^{-1} \sum_j n_j \, \bar{y}_{jR}}$$

Unbiased under quasirandomization (MCAR within each class). Each respondent in class j is weighted by sample share n_j over respondent share r_j. Example: a 4-stratum cholesterol survey — the naive respondent mean implicitly under-weights the sickest, worst-response stratum (7% vs its true 12% share); the class-weighted mean corrects this to ȳ_wc = 236.15.

PROPENSITY SCORE SUFFICIENCY

$$M \perp Y \mid p(X)$$

$$\boxed{p(X) = \Pr(\text{respond} \mid X)}$$

Proved sufficient by a 4-line tower-rule argument under MAR. A single logistic regression replaces sparse weighting-class cells and handles continuous covariates natively. Bucketed use (5–6 strata) is robust to mild misspecification; direct use w_i=1/p̂(x) leans fully on a correctly specified model.

Recovering Data, Risking Coherence

利用可能ケース分析— データを活かすほど整合性が崩れる

RANGE-SAFE PAIRWISE CORRELATION

$$r_{jk} \in [-1,\ 1]$$

$$\boxed{r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}\, s_{kk}}}}$$

All computed on the SAME jointly-observed base I_jk. With 20 variables each 10% independently missing, only $0.9^{20} \approx 12\%$ of rows survive complete-case analysis. Available-case analysis uses every observed pair instead — restricting covariance AND variances to the same joint base keeps each correlation individually valid.

THE THREE-VARIABLE PARADOX

$$r_{12} = 1, \quad r_{13} = 1, \quad r_{23} = -1$$

$$\boxed{\{r_{jk}\} \text{ individually valid } \Rightarrow\ \Sigma \text{ not psd}}$$

Each pair is jointly observed on a different subset of rows — no single row observes all three variables together, so nothing anchors the three pairwise estimates to one consistent dataset. Regression needs a positive-definite covariance matrix: this can break downstream analysis entirely.

Absorbing Bias Into the Weight

重み付き一般化推定方程式（GEE）

NAIVE COMPLETE-CASE GEE

$$M \perp (Y, Z) \mid X$$

$$\boxed{\sum_{i \in CC} D_i(x_i, \beta)[y_i - g(x_i, \beta)] = 0}$$

Sum over complete cases only. Consistent only if missingness may depend on covariates X already in the mean model, but not on the outcome Y or an auxiliary variable Z (e.g. a symptom linked to dropout).

WEIGHTED GEE (ROBINS ET AL., 1995)

$$w_i = 1/\hat{p}(x, z)$$

$$\boxed{\sum_{i \in CC} w_i(\hat{\alpha})\, D_i(x_i, \beta)[y_i - g(x_i, \beta)] = 0}$$

Weighting each complete case by estimated inverse propensity relaxes the condition to: missingness may ALSO depend on Z, provided Z enters the propensity model — algebraically the same cancellation trick as the odds-ratio proof.

Deletion & Weighting, At a Glance

4つのアドホック手法の比較

Every method here either throws away information or reweights it. None fixes bias from a mechanism that depends on the missing value itself — that requires an explicit model.

Method	What It Does	Unbiased When...	Cost
Complete-Case	Keeps only fully-observed rows	Mechanism is MCAR (or CC-regression: M depends on X only, not Y|X)	$\Delta_{CC} = \frac{n-r}{r}$ (varies by estimand)
Weighted Complete-Case	Reweights survivors by n_j/r_j per class	Quasirandomization: MCAR within each weighting class	$\frac{\sigma^2}{r}(1 + cv^2(w_i))$ variance
Propensity-Weighted	Reweights by 1/p̂(X), one logistic model	MAR: Pr(M|X,Y) = Pr(M|X); model correctly specified	Handles sparsity & continuous covariates; direct weights can be unstable
Available-Case	Uses every observed value/pair per variable	Same conditions as CC, applied per-pair instead of per-row	Pairwise correlations can combine into a non-PSD matrix
Weighted GEE	Weights each complete case's estimating equation	Pr(M|X,Y,Z) = Pr(M|X,Z); propensity model correct	Needs Taylor-series variance correction for estimated α̂

Mean Imputation: Unconditional vs Conditional

平均値代入 — 全体平均からセル内平均へ

UNCONDITIONAL MEAN

$$Y_i^* = \bar{Y}_{\text{obs}} \quad \text{for all missing } i$$

$$\boxed{s^2_{\text{filled}} = s^2_{\text{obs}} \frac{n_{\text{obs}} - 1}{n - 1} \approx (1 - \lambda) s^2_{\text{obs}}}$$

Every gap gets the same single number, so the completed data are artificially compressed: variance, covariance, and correlation all shrink. Preserves the mean under MCAR, but distorts almost everything else. Generally not recommended.

CONDITIONAL MEAN (WITHIN CELLS)

$$Y_i^* = \bar{Y}_{hR} \quad \text{for } i \in h$$

$$\boxed{\text{Bias}(\bar{Y}_{CM}) = \frac{1}{n} \sum_h m_h (\mu_{hR} - \mu_{hM}); \quad 0 \Leftrightarrow \mu_{hR} = \mu_{hM}}$$

Groups units by observed covariates (age × sex × region...), then imputes the respondent mean within each cell. Unbiased once respondents and non-respondents are comparable inside a cell — but still imputes one value per cell, so within-cell variance still shrinks.

Regression Imputation & Buck's Method

回帰代入からバック法へ— 共分散行列を使った一般化

REGRESSION IMPUTATION

$$\hat{Y}_i = X_i\hat{\beta}$$

$$\boxed{Y_i = X_i\beta + \varepsilon_i \quad \Rightarrow \quad Y_i^* = X_i\hat{\beta} \ \ (\varepsilon_i \text{ discarded})}$$

Generalizes conditional-mean imputation to continuous covariates. But every imputed point sits exactly on the regression line — real data scatter around it — so residual variance is systematically underestimated.

BUCK'S METHOD (GENERAL PATTERNS)

$$Y^*_{\text{mis}} = \hat{\mu}_{\text{mis}} + \hat{\Sigma}_{mo}\hat{\Sigma}_{oo}^{-1}(Y_{\text{obs}} - \hat{\mu}_{\text{obs}})$$

$$\boxed{E(s_{jj,\,\text{Buck}}) \approx \sigma_{jj} - \frac{1}{n}\sum_i \sigma_{jj\,|\,\text{obs},\,i}}$$

Extends regression imputation to ANY missing-data pattern at once, using one complete-case mean vector and covariance matrix — reused per pattern via the sweep operator, not refit per row. Efficient, but still a conditional-mean method underneath.

From a Mean to a Draw

確率的回帰代入 — 平均ではなく「もっともらしい値」を引く

Mean-based imputations (unconditional, conditional, regression, Buck's) all fill the center of a predictive distribution. Real missing values scatter around that center.

FORMULA

$$Y_i^* = X_i\hat{\beta} + z_i, \quad z_i \sim N(0, \hat{\sigma}^2_{\text{res}})$$

KEY TAKEAWAY

Regression gives the conditional mean. Stochastic regression gives a conditional draw.

Adding random residual noise restores realistic variability. For binary outcomes the same idea gives Y*_i ~ Bernoulli(p̂_i) instead of rounding a probability.

WHY IT MATTERS

This is the pivot point of the entire chapter.

Every method after this slide — hot deck, PMM, multiple imputation — is some version of 'impute a draw, not a mean'. The four-method comparison on the next slide proves it formally.

Umean, Udraw, Cmean, Cdraw

二変量正規分布における4手法の比較 — バイアスの数式

Umean · Unconditional

$$\mathrm{Bias}(\sigma_{22}) = -\lambda\sigma_{22}$$

Ignores Y1 entirely. Preserves the mean, destroys variance and the regression relationship.

Udraw · Marginal Draw

$$\mathrm{Bias}(\beta_{12}) \neq 0$$

Preserves marginal variance of Y2, but damages the Y1–Y2 relationship since it ignores Y1.

Cmean · Conditional

$$\mathrm{Bias}(\sigma_{22}) = -\lambda(1-\rho^2)\sigma_{22}$$

Keeps the regression slope, but residual scatter is still missing — variance still shrinks.

Cdraw · Conditional Draw

$$\mathrm{Bias} = 0 \ \ \forall\ \mu, \sigma_{22}, \beta_{21}, \beta_{12}$$

The only method among all four with zero bias across every listed parameter — proof that draws beat means once you also condition on what you know.

Efficiency Tradeoff

$$+\frac{\lambda(1-\lambda)(1-\rho^2)\sigma_{22}}{n}$$

Cdraw's only cost: extra sampling variance on the mean. Realism has a small, honest price.

Hot Deck: Copying a Real Respondent

ホットデック法— 平均ではなく実在する値をコピーする

SIMPLE RANDOM HOT DECK

$$Y_i^* = Y_J, \quad J \sim \mathrm{Unif}(\text{respondents})$$

$$\boxed{\mathrm{Var}(\bar{Y}_{HD1}) = \left(\tfrac{1}{r} - \tfrac{1}{N}\right) S_y^2 + \left(1 - \tfrac{1}{r}\right)\left(1 - \tfrac{r}{n}\right)\tfrac{S_y^2}{n}}$$

E(Ȳ_HD1) = Ȳ under MCAR — unbiased. The second variance term is an honest extra cost of donor sampling: donor counts (H1,...,Hr) follow a Multinomial(n−r; 1/r,...,1/r).

HOT DECK WITHIN ADJUSTMENT CELLS

$$Y_i^* = Y_J, \quad J \in D_i = \{j : R_j = 1,\ j \in h\}$$

$$\boxed{E(Y_i^* \mid Y_{\text{obs}}) = \bar{Y}_{hR}}$$

Matches cell-mean imputation in expectation — but every imputed value is a real observed value, so realistic scatter survives instead of being flattened to an average.

Nearest-Neighbor (KNN) Hot Deck

最近傍法ホットデック— 距離に基づくドナー選択

Donors are selected from the nearest respondents in covariate space, using a distance d(i,j).

DISTANCE METRICS

Mahalanobis & Predictive-Mean Distance

$$d(i,j) = (X_i - X_j)^T S_{XX}^{-1} (X_i - X_j)$$

$$d(i,j) = [\hat{Y}(X_i) - \hat{Y}(X_j)]^2$$

Other options include maximum-deviation and plain Euclidean distance.

THE SCALING PROBLEM

Large-scale variables silently dominate

Example: income spans 0 to 2,000,000 while age spans 18 to 80. Raw Euclidean distance mostly measures income, not age.

$$Z = \frac{X - \bar{X}}{s_X}$$

Standardize first, or use domain weights, anchor groups, Mahalanobis distance, or predictive mean matching instead.

Sequential Hot Deck

順序型ホットデック— アンカーグループと並べ替え

Form an exact-match anchor group A_i (e.g. region × industry), sort records inside it by a variable related to the missing value, then choose a nearby donor in that sorted order.

DONOR WINDOW

$$D_i(L) = \{\, j : R_j = 1,\ A_j = A_i,\ |\mathrm{rank}(j) - \mathrm{rank}(i)| \le L \,\}$$

REAL EXAMPLE: CONSTRUCTION SURVEY

Anchors + sort + distance

Post-strata by province × SIC × SWI; sort within each by gross business income; take the nearest 5 donors either side; distance $d(i,j)=|\ln(TEXP_i)-\ln(TEXP_j)|$. A usage penalty $d^*=d+\gamma u_j$ stops one donor from being reused too often.

EDIT CONSTRAINTS

Respecting logical totals

If $x+y \le z$ must hold and candidate i has z_i but not x_i, y_i, copy the donor's PATTERN, not its raw values: $x^*_i=(x_j/z_j)z_i$, $y^*_i=(y_j/z_j)z_i$ — transfers shape while respecting the candidate's own total.

Cold Deck Imputation

コールドデック法— 外部・過去のデータを使う

FORMULA & BIAS · unbiased iff C(X_i) = m_M(X_i)

$$Y_{CD,\,i} = R_i Y_i + (1 - R_i) C(X_i); \quad \mathrm{Bias}(\bar{Y}_{CD}) \approx \frac{1}{n} \sum_i [C(X_i) - m_M(X_i)]$$

Feature	Sequential Hot Deck	Cold Deck
Source	Current sample donors	External / old source
Value copied	Actual current respondent value	Lookup / historical value
Adapts to current sample	Yes	Often weakly
Main risk	Bad matching or donor overuse	External value is wrong or outdated
Uncertainty	Still understated in single imputation	Often severely understated

Substitution & Repeated-Measures Methods

代替法とLOCF・行列効果法

SUBSTITUTION

$$Y_i^* = Y_j \quad (\text{fresh, unselected unit})$$

Bias = μ_S − μ_M. Makes fieldwork easier but does not remove nonresponse bias — it merely hides it inside a replacement unit that is itself a respondent, not a true nonrespondent.

LOCF (DROPOUT)

$$\hat{Y}_{it} = Y_{i,\,k-1} \quad \text{for } t = k, \ldots, K$$

Last Observation Carried Forward assumes the outcome stays unchanged after a subject drops out — simple, but often unrealistic and prone to bias in longitudinal studies.

ROW + COLUMN FIT

$$\hat{Y}_{it} = Y_{lt} + (\bar{Y}_{\text{obs},\,i} - \bar{Y}_{\text{obs},\,l})$$

Borrows donor l's residual deviation, adjusted for subject i's own mean. A multiplicative variant uses Ŷ_it = Y_lt·(Ȳ_obs,i/Ȳ_obs,l) when effects scale rather than shift.

Predictive Mean Matching (PMM)

予測平均マッチング— 回帰とホットデックの融合

A model defines similarity; an actual observed value gets imputed. The hybrid that bridges explicit and implicit modeling.

1

Fit a model

$\hat{Y} = \hat{m}(X)$ using observed cases only.

2

Predict for everyone

Compute $\hat{m}(X)$ for both observed and missing cases.

3

Find close matches

For missing case i, find observed j with small $|\hat{m}(X_i) - \hat{m}(X_j)|$.

4

Draw a donor

Randomly choose one donor from the closest candidates — not just the nearest.

5

Impute the real value

$Y^*_i = Y_J$ — the donor's actual observed value, never a predicted number.

KEY TAKEAWAY

PMM uses a model to define similarity but still imputes an actual observed value — regression structure with hot-deck realism, and no distributional assumption on the residual.

Mathematical Proof, or Empirical Claim?

どこまでが数式で証明され、どこからが経験則か

Most of these bias/variance claims are formal results under stated assumptions, not simulation folklore - except where the donor-selection algorithm is too complex for closed form.

Method	Mathematical Claim	Nature of Claim
Unconditional mean	Variance/covariance attenuation factor (n_obs-1)/(n-1)	**Formal under MCAR**
Conditional cell mean	Bias = weighted respondent/non-respondent gap inside cells	**Formal decomposition**
Regression imputation	Removes residual variance by construction (imputes conditional mean)	**Model-based**
Buck's method	Variance/covariance underestimated by residual conditional-variance terms	**Formal under assumptions**
Cdraw (bivariate normal)	Bias exactly zero for μ, σ22, β21, β12	**Formal, worked example**
Simple random hot deck	Donor counts ~ Multinomial; closed-form variance available	**Formal**
Sequential / KNN hot deck	Bias depends on matching quality and distance rule chosen	**Less complete theory**
Cold deck	Bias depends entirely on correctness of the external source	**Obvious, but source-dependent**

Testing the Theory Against Real Numbers

理論を実際のシミュレーションデータで検証する

DESIGN

Six missingness scenarios, one linear model

A single regression Y~X is estimated after inducing ~30% missingness in one variable at a time, under all three mechanisms: MCAR, MAR (driven by a measured covariate Z or by the other variable), and MNAR (driven by the variable's own unobserved value).

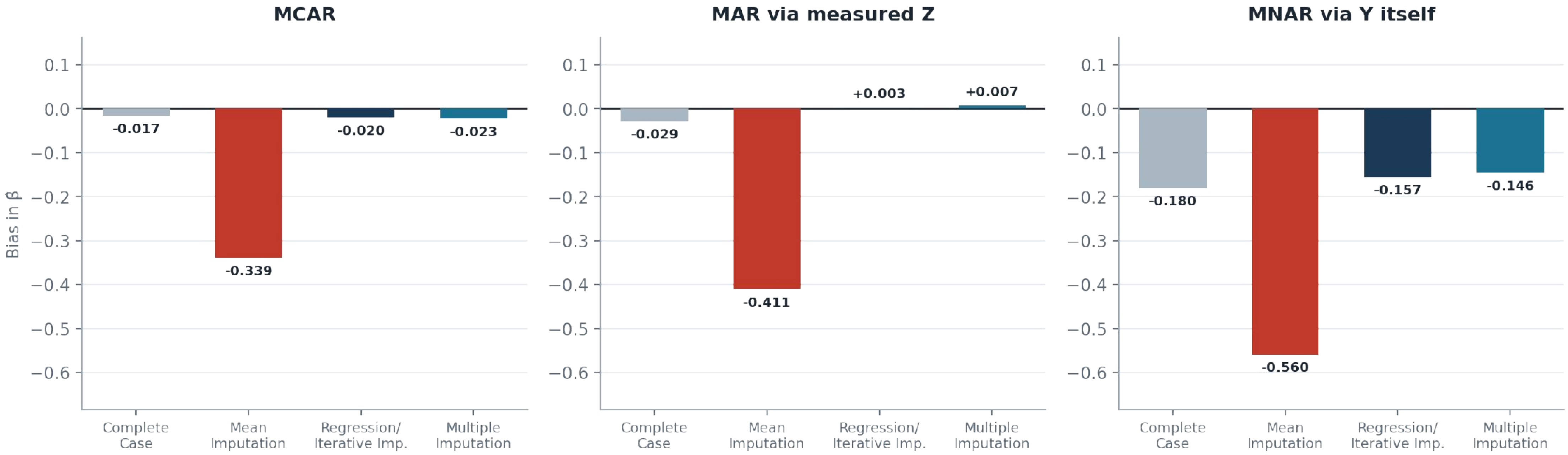
METHODS COMPARED

Complete case · Mean · Regression/iterative · Multiple imputation

Every scenario is run through all four methods and compared against the known true $\beta=1.10$, recording $\hat{\beta}$, Bias = $\hat{\beta}-\beta$, and the standard error - exactly the quantities the theory chapters make formal predictions about.

Bias by Method: Y Is Missing

従属変数Y が欠測している場合のバイアス比較

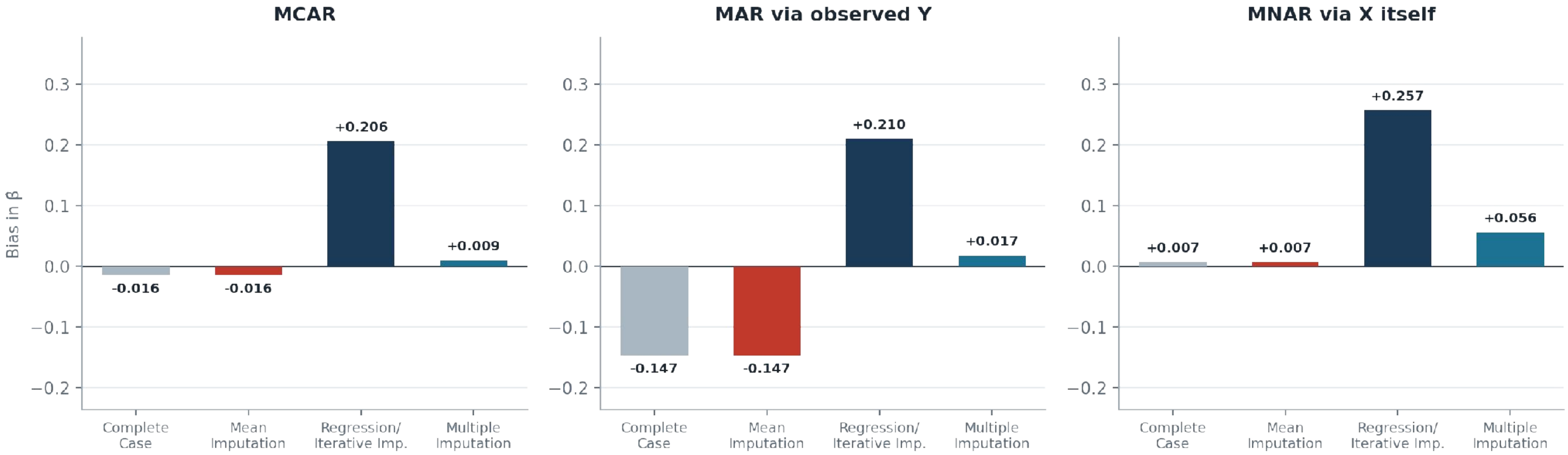


READING THE CHART

Mean imputation's bias explodes as the mechanism worsens (−0.34 → −0.41 → −0.56). Multiple imputation tracks the theory's prediction closely and stays smallest or near-smallest throughout.

Bias by Method: X Is Missing

説明変数X が欠測している場合のバイアス比較



A COUNTERINTUITIVE RESULT

Regression/iterative imputation - the strongest method for Y-missing - becomes the WORST here (+0.21 to +0.26 bias). Imputing a predictor from itself manufactures correlation that was not there.

Key Empirical Findings

実験からわかった3つの教訓

FINDING 1

Mean imputation is unsafe under any mechanism

Bias worsens monotonically from MCAR (−0.34) to MNAR (−0.56) when Y is missing - exactly the attenuation the textbook formula $(1-\lambda)s^2\_obs$ predicts, at real magnitude.

FINDING 2

Imputing a predictor is not symmetric with imputing an outcome

Regression imputation excels for Y-missing but is the single worst method for X-missing - it manufactures a correlation between X and its own predictors that inflates $\hat{\beta}$.

FINDING 3

Multiple imputation is the most consistently reliable

Smallest or near-smallest |bias| in 5 of 6 scenarios, with SE of comparable size to the other methods - confirming Rubin's Rules honestly price in imputation uncertainty without overcorrecting.

PART 2

Cultural Experience

文化体験

Three cities that shaped my time in Japan as an exchange student

Tokyo - The Capital and Chaos

東京

東京滞在中に、浅草、秋葉原、銀座、上野、新宿、渋谷を訪れる機会がありました。街の活気や賑わいは、言葉では十分に表すことができません。一日を通して多くのことが同時に起こっており、これほど多くの観光客が集まっている様子を見るのも初めてでした。

これまで訪れた国々の中でも、東京の公共インフラは特に優れていると感じました。特に、地下鉄やJR線を含む広範な鉄道網には大変感銘を受けました。また、新幹線に乗車する機会にも恵まれ、私にとって非常に思い出深い経験となりました。

PART 2 · CULTURAL EXPERIENCE

Osaka - Hidden Japan, Mini Tokyo

大阪

大阪は、活気、温かさ、そして賑やかさに満ちた、日本のまた違った一面を表していました。京都の伝統的な雰囲気や、東京の速く整った都市の姿とは異なり、大阪はより親しみやすく、表現豊かで、人とのつながりを感じられる街でした。特に道頓堀周辺の賑やかな通りは、大阪の有名な食文化と明るい雰囲気をよく表していました。交換留学生として、大阪では食べ物、会話、ユーモア、そして人々の日常的な温かさを通して、日本のより気さくで活気ある文化を感じることができました。

Kyoto - Where the Past meets the Present

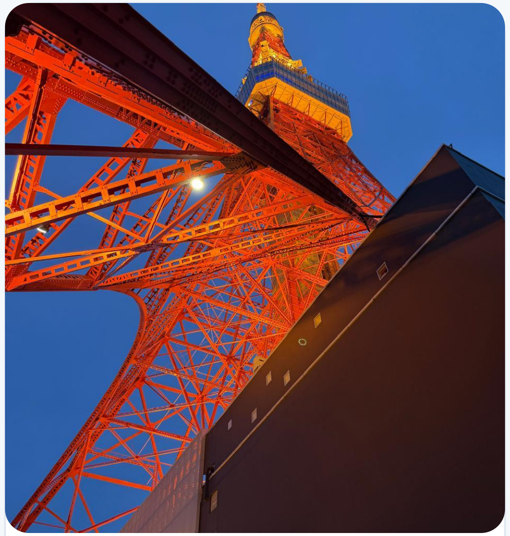
京都

京都は、日本の真の文化的中心地を象徴しており、過去と現在が美しく融合している場所でした。特に伏見稲荷大社をはじめとする神社は本当に素晴らしく、その伝統的な建築は、東京や大阪の近代的な景観とははっきりと対照的でした。交換留学生として、私は日本の礼儀作法や社会的な規律を間近で感じることができました。特に、人々が優雅に規則を守る姿や、お辞儀の文化に表れる深い敬意がとても印象的でした。

PART 2 · CULTURAL EXPERIENCE

Exchange Journey Gallery

思い出のアルバム · A visual record of cities, culture, and connections



Tokyo 東京

Osaka 大阪

Kyoto 京都

Cultural Experiences 文化体験

Thank You / ありがとうございました

Nagasaki University

長崎大学

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Hiromi Collins Sensei

コリンズ・ヒロミ先生

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Questions are welcome — 質問を歓迎します
GitHub Link — MadhavGaur123/Nagasaki-University-